

FOX_Checklist

What this is: every FoxByte feature and its sub-features — built or not, and how it was tested — so it's clear where we are and what's next. How each feature works is in `FOX_Feature_Implemented.pdf` (same ids).

Last updated: 22 September 2026 · **Reflects:** `main @ 2778762` (v1.0.1, exact repository path PR #1) + the FoxByte rename and brand indirection (`chore/foxbyte`)

Source: `docs/FOX_Checklist.html` → `make feature-doc` → this PDF

☒ Done & tested — automated tests (unit, VM integration or CI)

☒ Done — tested by hand only

☐ Partial — gaps, or not verified end to end

☒ Known issue

☐ Not started

Sub-features: ☒ built ☐ partly ☐ not built yet ☒ broken · "VM §n" = section of `make integration` ; "v2 §n" = `make integration-v2` ; "update" = `make integration-update`

Where we are

Area	Features	Done & tested	Done (manual)	Partial	Known issue	Not started	Progress
A · Install & setup	9	4	3	2	·	·	78%
B · CLI & services	2	2	·	·	·	·	100%
C · Branching & storage	8	7	·	1	·	·	88%
D · Durability	6	5	·	1	·	·	83%
E · Security foundations	3	3	·	·	·	·	100%
F · SQL gateway	7	6	1	·	·	·	100%
G · Accounts & auth	6	5	·	1	·	·	83%
H · Blackbox (record of schema changes)	18	17	·	1	·	·	94%
I · Import & migration	13	5	8	·	·	·	100%
J · ETL pipelines (experimental)	4	2	2	·	·	·	100%
K · APIs, SDKs & agents	6	4	2	·	·	·	100%
L · Web console	11	1	10	·	·	·	100%
M · Build, CI & releases	7	6	1	·	·	·	100%
Total (features built or started)	100	67	27	6	·	·	94%
N · Not started (roadmap)	11	1	·	·	·	10	9%

Storage engine — not implemented, by design

FoxByte does **not** have its own storage engine. Data is stored by **PostgreSQL's own engine** (heap tables, B-tree indexes, write-ahead log) running on local disk inside the stock `postgres:18-bookworm` image plus `wal-g` (`postgres:16-bookworm` for an install created before 21 Sep 2026). This was a locked architecture decision: a Neon-style engine that separates compute from storage over the network adds latency to every transaction, and transaction speed must never be compromised.

What **is** built is a storage *layer* around Postgres: copy-on-write branch drivers — ZFS on macOS/Linux, btrfs on Windows — behind one interface in `internal/branch/storage.go` (C1), quotas (C6), capacity reporting (C7), and durability to object storage with `wal-g` + MinIO: WAL archiving, base backups, point-in-time restore (D1–D3). Networked storage disaggregation is listed in the README roadmap but not started (see "Not started").

Suggested next steps

- Watch the first nightly run** (Actions → nightly): it is the first time the suites run on GitHub's runners, and it says whether their kernel carries the ZFS module (M2).
- Cut the first signed release** and check `gh attestation verify` and `fox update` against it (M4, A9).

3. Run on real machines what the throwaway VM cannot: `fox update` across a real release on macOS and Windows (A9); on Windows, the `zstd` distro image's unpack (M5), the `btrfs` holding area of `fox pg upgrade` (D6), and an export's copy to the host (D5).
4. Make the `postgres-wal-g` image package public, so installs pull it instead of building it for a few minutes (C8).
5. Run the VM integration suites in CI on every pull request (M2).
6. Add automated tests where there are none: imports (I1–I3, I7), SDKs (K3, K4), web UI (L*).
7. HA beyond the demo: automatic failover and fencing (D4).

A Install & setup

7 of 9 done

ID	Feature · sub-features	Tested by
 A1 One-line installer (macOS / Linux)	✓ Detects OS / CPU, downloads the release binary ✓ Installs to the prefix (sudo only if needed) ✓ Stages the Linux engine for the VM on macOS ✓ <code>FOX_VERSION</code>, <code>FOX_REPO</code>, <code>FOX_DIST</code>, <code>FOX_PREFIX</code> ✓ Verifies every download against <code>SHA256SUMS</code>; refuses what it can't check ✓ Release signature checked (OpenSSL 3; warns with LibreSSL)	Unit (<code>test_install_sh.sh</code> , CI); manual
 A2 Windows installer (install.ps1)	✓ Installs WSL, elevates, resumes after reboot ✓ Downloads fox.exe, engine, image build context ✓ Ubuntu rootfs verified against Ubuntu's checksums ✓ Adds to PATH and runs <code>fox setup</code> ✓ Prefers the prebuilt distro image, falls back to the rootfs ✓ Verifies FoxByte's own downloads ✓ Run on a real Windows machine (distro image downloaded and verified)	Pester (Windows CI); Windows run 16 Sep
 A3 fox setup on macOS (Lima)	✓ Creates / starts the Lima VM ✓ Installs Docker and ZFS ✓ Reinstalls the engine on every run ✓ Starts the stack ✓ Repairs an interrupted provisioning on the next run (dpkg interrupted, half-installed packages) ✓ Image builds from the copy inside fox when it cannot be pulled (C8)	Unit; run against a deliberately broken VM, 21 Sep
 A4 fox setup on Windows (WSL2 + btrfs)	✓ Imports a dedicated distro with systemd ✓ Docker, btrfs and a storage mount unit ✓ Quiet steps, detail in install.log ✓ Waits for systemd on a cold distro ✓ Preloaded images tagged as the engine runs them ✓ Summary with API key and <code>sslmode</code> ✓ Fresh install from v0.8.6 on Windows: preload used, no apt, key in the summary	Unit (<code>host_wsl_test.go</code>); Windows run 16 Sep
 A5 Linux native	✓ Engine runs in-process, no VM ✓ <code>fox setup</code> explains nothing is needed ✓ Services listen on loopback by default (<code>FOX_LISTEN</code> to expose)	VM integration (all suites run on Linux)
 A6 Command forwarding & host file streaming	✓ macOS → Lima VM, starts the VM if stopped ✓ Windows → WSL distro ✓ <code>fox import</code> streams a host file over stdin □ Stream <code>pipeline run spec</code> files too	Unit (<code>TestImportLocalFile</code> , <code>TestWSLargs</code>)
 A7 Always-latest engine binary	✓ Downloads the newest engine into the cache (ELF check) ✓ <code>FOX_NO_REFRESH</code>, <code>FOX_VERSION</code> ✓ Verified against <code>SHA256SUMS</code>; keeps the staged engine if not ✓ Seen verifying on a real Windows install	Unit (<code>host_test.go</code> , <code>update_test.go</code>); Windows run 16 Sep
 A8 New-release notice on fox start	✓ Background check, 1.5 s limit, printed last ✓ Only a promoted, newer, complete release ✓ One request per start; a found release remembered 6 h, never "up to date" ✓ Off switch; never on dev builds; silent offline	Unit; update §1, §1b, §2, §3, §8; real Mac check
 A9 fox update	✓ Downloads and verifies everything before changing anything ✓ Stops servers, installs engine, restarts old one on failure ✓ up, Blackbox upgrade, <code>start</code> on the new engine ✓ Replaces host <code>fox</code>, keeps previous binaries ✓ <code>--check</code>, <code>--yes</code>, <code>--version</code> ✓ Linux path verified end to end ✓ macOS path run against real releases (v0.8.3 → v0.8.4 → v0.8.5) □ Windows path run against a real release (deferred) ✓ Release signatures (Ed25519 over <code>SHA256SUMS</code>)	Unit; update §4–§7; a real Mac run

B CLI & services

2 of 2 done

ID	Feature · sub-features	Tested by
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✓	B1 The fox command set	✓ Stack: setup, start, stop, up, down, status, logs, psql, update ✓ Branching, backup / restore, HA ✓ Blackbox, policy, impact ✓ Import, pipelines ✓ backup export / restore, pg status / upgrade	✓ Users, API keys, admin ✓ <code>fox vm status / fox vm shell</code> (Lima, WSL; Linux says there is no VM) ✓ <code>branch create --from</code> ✓ <code>fox uninstall</code> removes it all, retired installs included	Unit (<code>main_test.go</code> , <code>update_test.go</code>); VM suites
✓	B2 Background services (daemon)	✓ Control plane :8080, gateway :6432, agent API :8088 ✓ Detached, pidfiles and logs in <code>~/ .fox</code>	✓ Stop (TERM then KILL), status, logs □ Health check / restart on crash	Unit (<code>TestStopTerminatesProcess</code>); VM §1

C Branching & storage

7 of 8 done

ID	Feature · sub-features	Tested by
✓ C1	Storage drivers (ZFS / btrfs) ✓ ZFS snapshot + clone per branch (macOS VM, Linux) ✓ btrfs subvolume snapshots (Windows) ✓ Pool / filesystem created on a loopback file	Unit; VM §2, §2b (ZFS); btrfs manual on Windows
✓ C2	Branch lifecycle ✓ Create, list, delete ✓ Reset to parent ✓ Suspend / resume ✓ Recreate a container from its surviving data	Unit; VM §2, §2b, §4
✓ C3	Protection of main ✓ Delete and reset refused	Unit; VM §4, §6
✓ C4	Private networking (no host ports) ✓ Branch databases reachable only through the gateway ✓ Published ports (object store, restore, debug) on loopback only; standby unpublished	Unit (<code>TestParsePublishedPort</code> , <code>TestDSN</code>); VM §1c
✓ C5	Scale-to-zero & wake on connect ✓ Idle branches suspended automatically ✓ Next connection wakes the branch (~0.3–0.7 s)	VM §7
🔥 C6	Storage quotas ✓ Space reserved for main (ZFS) 🔥 Per-branch cap (opt-in)	Manual
✓ C7	Capacity & status ✓ Services, main health, backups, branches ✓ Storage used / available	VM §1
✓ C8	Provisioning & image pull ✓ Docker, storage pool and image prepared on up	Unit; VM §11b
		✓ Pulls from GHCR, builds locally as fallback ✓ The build context is built into fox, so the fallback works on any install

D Durability

5 of 6 done

ID	Feature · sub-features	Tested by
✓ D1	WAL archiving to object storage ✓ wal-g ships main's WAL to MinIO, off the commit path ✓ MinIO images pinned on quay.io ✓ Branches are disposable by design (export one that matters)	VM §3; VM §11c
		✓ Remote S3 target (<code>fox backup target set</code>), keys in 0600 env files ✓ Object Lock reported; tested on MinIO □ Tested against AWS S3

✓	D2 Base backups	✓ Create and list ✓ Follow the primary after a failover; listing works with main stopped ✓ Read-only listing: <code>GET /api/backups</code> and a Dashboard panel	✓ Schedule and retention (<code>FOX_BACKUP_INTERVAL</code> , <code>FOX_BACKUP_RETAIN</code>) ✓ A first backup at install ✓ Age shown and warned in <code>fox status</code> and the status API □ Automated test restore	VM §3, §3b, §6; VM §11c
✓	D3 Point-in-time restore	✓ Restore to a timestamp or latest on :5433 ✓ Per-install credentials ✓ Starts from the newest base backup that precedes the target	✓ A point no backup precedes is refused, naming the oldest	Unit; VM §3, §3b, §6; v2 §1a
🔥	D4 HA standby & failover	✓ Streaming hot standby ✓ Manual failover, <code>main</code> rerouted ✓ Nothing deletes or stops the promoted standby ✓ <code>fox up</code> brings it back, not the old main	✓ <code>fox ha fallback</code>, keeping every write ✓ The promoted standby archives WAL, so backups and PITR cover the failover □ Automatic failover, fencing, multi-host	Unit; VM §6, §7
✓	D5 Portable export & restore	✓ Every non-agent branch: <code>pg_dump</code>, its roles, the Blackbox head; SHA256SUMS over all ✓ Copied out of the VM and verified on the host; a copy that fails is deleted ✓ Damaged, incomplete or path-escaping archives refused	✓ Restore into a new branch, same or newer major; Blackbox checked exact 🔥 Host copy on macOS/Windows unit-tested only	Unit; VM pg-upgrade §1
✓	D6 PostgreSQL major upgrade	✓ <code>fox pg status</code> : the major of the install and of each branch ✓ Plan with every change crossed, found-in-your-data first; refusals say what to do ✓ Asks, exports, then reloads every carried branch on the new major	✓ Data, roles, API keys and every Blackbox entry come across ✓ Old databases held outside the branch namespace; rollback and finalize ✓ Each upgraded cluster archives WAL under its own prefix 🔥 btrfs (Windows) holding area not yet run □ Carry agent branches; upgrade with HA on	Unit; VM pg-upgrade (72 / 72)

E Security foundations 3 of 3 done

ID	Feature · sub-features	Tested by
 E1	Per-install generated secrets ✓ Generated Postgres and MinIO passwords ✓ Used by restore too ✓ Each login role has its own derived password	Unit (indirect); v2 §1a; VM §1d
 E2	TLS everywhere ✓ Self-signed TLS on all three services, zero setup	Unit (<code>TestReadStartupUpgradesTLS</code>)
 E3	Keys are made by a person, not by start ✓ Start creates no account and no key, and caches none on disk ✓ The banner shows the connection string and how to make a key ✓ The first account created gets the admin grant on main	Integration §1

F SQL gateway 7 of 7 done

ID	Feature · sub-features	Tested by
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✓	F1 Routing by database name	✓ One endpoint :6432, database name = branch □ Connection pooling, rate limiting	Unit; VM §1, §6
✓	F2 API-key authentication	✓ key_ key as the Postgres password ✓ Startup deadline and connection cap ✓ Bad key rejected with a Postgres error ✓ An account key opens main, its own branches, or any for an admin	VM §1
✓	F3 Backend login (SCRAM)	✓ Cleartext, md5, SCRAM-SHA-256 □ Unit test for SCRAM ✓ Clients never see the database password	VM §1, §6, §8 (exercised)
✓	F4 Per-user Postgres login roles	✓ Role per user, actor can't be forged ✓ Own password per role ✓ Permissions of non-superuser db_client	Manual
✓	F5 Attribution session settings	✓ Actor kind, branch, session id per connection ○ Kind / branch / session client-settable	VM §8
✓	F6 Ctrl-C (CancelRequest) forwarding	✓ Cancel a running query through the gateway	Unit
✓	F7 Errors & insecure build tag	✓ Real Postgres error responses ✓ No-auth bypass impossible in release builds	Unit (TestSendError)

G Accounts & auth 5 of 6 done

ID	Feature · sub-features	Tested by
✓	G1 Auth store (SQLite) ✓ Users, keys, sessions, OAuth ids, pipelines ✓ Safe under concurrent use (WAL, busy timeout), including a new store created by three services at once □ Roles / permissions, multi-tenant projects	Unit (TestVerifyKeyConcurrent , TestConcurrentFirstOpen); VM §1
✓	G2 Email & password users ✓ Create account, log in ✓ Password change (signs out other sessions); admin reset at the CLI □ Self-service reset by email ✓ List and delete users (admins) ✓ Emails validated; at most 63 bytes (a role name)	Unit; VM §1; VM §1d
✓	G3 API keys ✓ Create (shown once), list, revoke ○ Revoking an unknown id reports success ✓ Bearer / X-API-Key and gateway password	Unit; VM §1
✓	G4 Web sessions & signup control ✓ httpOnly session cookie ✓ Failed sign-ins throttled ✓ Sign-up closed by default (FOX_SIGNUP=open)	Unit (TestSessions)
○	G5 OAuth (GitHub / Google) ✓ Sign-in flows built □ Tested with a real OAuth app ✓ Defaults match what the engine serves (https://localhost:8080 ; web origin = public URL) ✓ Only a provider-verified email joins or creates an account; closed sign-up respected ✓ Session cookie Secure / SameSite follow the origin ✓ State cookie one-use, Secure , SameSite=Lax	Unit; VM §1b
✓	G6 Security log ✓ Sign-ins, failures, keys, passwords, accounts, admin grants, refusals ✓ fox_audit , verify , GET /api/audit ✓ Hash-chained, append-only, anchored and signed □ Included in fox backup export	Unit; VM §1d

H Blackbox (record of schema changes) 17 of 18 done

ID	Feature · sub-features	Tested by
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✓	H1 DDL recording	✓ Who, tool, branch, statement, status, risk ✓ Inherited by every branch ✓ A restart records nothing: the install script's own statements are not captured ❑ Role / database changes, some dropped object types	✓ Agents' UPDATE and DELETE recorded; TRUNCATE by anyone ✓ Triggers on every table, protected from drop/disable/rename	VM §8; v2 §9; v2 §10
🔗	H2 Hash chain, append-only & verify	✓ Rows can't be edited or deleted normally ✓ <code>fox ledger verify</code> 🔗 Superuser rewrite caught only via anchors (H10–H12)	✓ <code>bb.policy</code> changes recorded (append-only, not chained)	VM §8; v2 §0, §6b
✓	H3 Attribution	✓ Humans via their login role ✓ Agents via their own <code>agent-<id></code> role	✓ Kind derived from the role, so neither can pose as the other ✓ Tool and session are client-declared by design, and labelled so	Unit; VM §8; v2 §1c
✓	H4 Destructive-DDL guardrail	✓ DROP TABLE / DROP SCHEMA refused, attempt recorded ✓ Override only superusers / <code>db_admin</code>, use recorded	✓ "flag" and "allow" actions; unknown actions refused ✓ No hang when blocked after a write in the same transaction ✓ TRUNCATE blocked by default, recorded	VM §8; v2 §1b, §6b
✓	H5 Risk flags	✓ Type change, drop column flagged ✓ Judged on the running statement (no comments or literals)	✓ Automated test	Unit; v2 §6b
✓	H6 Blackbox access (CLI / REST / MCP / UI / SDK)	✓ CLI, REST with filters, MCP, web page, SDKs	❑ Per-user access control	v2 §0, §5
✓	H7 <code>fox ledger revert --to</code>	✓ Help and output say it's a point-in-time restore of main	✓ Points to H13 for a branch from before a change	Unit (help text); VM §3 (the restore)
✓	H8 Override permission (<code>db_admin</code>) & upgrade	✓ <code>fox admin grant revoke list</code> ✓ First user granted automatically	✓ <code>fox blackbox upgrade [--all]</code> ✓ Grant / revoke from the web (admins only)	v2 §1b, §6
✓	H9 Blackbox 2.0 capture	✓ Transaction id and WAL position per entry ✓ Task, tool-call hash, override use	✓ Fail-safe, append-only ✓ Kill switch not settable by clients	Unit; v2 §2
✓	H10 Checkpoints & anchors	✓ Merkle root in the database and in read-only anchor files ✓ On a schedule and on demand ✓ Published anchor format	✓ Copied to the backup target (S3 Object Lock) ✓ Anchors signed (Ed25519, key outside the databases)	Unit; v2 §3, §3b
✓	H11 <code>fox-verify</code> (independent verifier)	✓ Checks a live branch or an export ✓ Release binaries for every platform	✓ <code>--pubkey</code> checks signatures	Unit; v2 §3
✓	H12 Blackbox integrity & export	✓ INTACT / TAMPERED via CLI, REST, MCP ✓ JSON-lines export	❑ Streaming for very large ledgers	v2 §3
✓	H13 Branch from just before a Blackbox entry	✓ Restore + replay up to the change's transaction ✓ By transaction id or timestamp	✓ CLI, REST, MCP, web panel ✓ main untouched	Unit; v2 §4
✓	H14 Blackbox names & aliases	✓ <code>fox blackbox</code>, <code>/blackbox</code> routes, MCP and SDK names	✓ Every original name still works	Unit; v2 §5

✓	H15 Policy gate (warn / block)	✓ BBX02 warn / BBX01 block with JSON detail ✓ Blocked attempts recorded ✓ Per-rule <code>db_admin</code> override	✓ CLI, REST, MCP, web page; custom rules ✓ Patterns match the running statement ✓ TRUNCATE (its own guardrail, H4)	Unit; v2 §6, §6b
✓	H16 Agent provenance & execute_change	✓ Session, task, parent session on every agent change (also through the Gateway DSN) ✓ MCP <code>execute_change</code> with policy preview	✓ <code>fox blackbox sessions</code>, REST ⦿ Provenance is what the agent says (not verified)	Unit; v2 §7
✓	H17 Impact analysis	✓ Dependents (views, foreign keys, indexes, depth) ✓ Other branches that have the object ✓ Policy verdict and risk score	✓ CLI, REST, MCP, inside <code>execute_change</code> ⦿ Functions, types, schemas not recognised from SQL	Unit; v2 §8
✓	H18 Blackbox diff between branches	✓ Fork point, changes only on each side ✓ Objects changed on both	✓ CLI, REST, MCP □ Merge (diff only today)	Unit; v2 §8

I Import & migration 13 of 13 done

ID	Feature · sub-features	Tested by
✓ I1	From Postgres ✓ Schema and data via <code>pg_dump</code> ✓ Zero-table result reported as an error	□ Automated test Manual
✓ I2	From MySQL ≤5.7 / MariaDB ✓ pgloader with type mapping (multi-arch image)	□ Automated test Manual
✓ I3	From MySQL 8.x ✓ Native <code>mysqldump</code> path (<code>caching_sha2_auth</code>) ✓ Types, NULLs, unicode, PK	□ Other indexes, foreign keys, defaults □ Automated test Manual
✓ I4	From MongoDB ✓ Collection → typed table, exact name	✓ Nested objects / arrays kept as jsonb VM §9, §10
✓ I5	From a .sql dump ✓ Runs the dump into a new branch ✓ Fails listing every failed statement (instance kept)	✓ Same for Postgres imports (I1) ✓ Automated test Unit; VM §11
✓ I6	From JSON / NDJSON ✓ Columns inferred from top-level keys	✓ An invalid document fails the import, naming its element or line Unit; VM §10, §11
✓ I7	From CSV ✓ Table from the header, exact column names	□ Typed columns (all text today) Manual
✓ I8	Continuous replication & cutover ✓ Initial copy + streaming (logical replication) ✓ Cutover keeps data (CLI) ✓ Status and cutover from REST and the Import page	✓ A replicating branch is never auto-suspended ✓ Automated test (VM §5b) and driven in the browser Unit; VM §5b
✓ I9	Guard handling during import ✓ Guardrail paused, recording stays on	✓ Re-armed even after a crash Manual
✓ I10	Live progress (CLI & SSE) ✓ Live log and progress bar	⦿ <code>psql</code> subprocess output not streamed Manual
✓ I11	Import REST endpoints ✓ From a connection string or an uploaded file □ Target name validation	✓ Private, local and container sources for admins only; metadata refused Manual; unit; VM §1d

✓	I12 Web Import page	✓ Database and file panels with live console ✓ No longer claims a SQLite loader that doesn't exist	Manual
✓	I13 Exact identifier fidelity	✓ Case kept (<code>userId</code> , <code>leaseAiChats</code>) ⚠ Field order of JSON sources not kept	Unit; VM §10

J ETL pipelines (experimental)

4 of 4 done

ID	Feature · sub-features	Tested by
✓ J1	Pipeline engine ✓ Extract → raw schema → SQL models → tests, in a fresh branch ✓ <code>ref()</code> / <code>source()</code> templating □ Dependency graph □ Incremental models □ Scheduling ✓ A run replaces only a branch the pipeline made ✓ API sources are connection strings, never server paths	Unit; VM §9
✓ J2	Data-quality tests ✓ <code>not_null</code>, <code>unique</code>, <code>accepted_values</code>, <code>row_count_min</code>, <code>custom</code> ✓ Failure → exit 1, data kept	Unit; VM §9
✓ J3	Stored pipelines & run history ✓ Saved per user, run over the API, history □ Read a run's log □ Mark runs whose process died	Manual
✓ J4	Pipeline editor (web) ✓ Create, edit, run, results, history □ Confirm before delete	Manual

K APIs, SDKs & agents

6 of 6 done

ID	Feature · sub-features	Tested by
✓ K1	Control-plane REST API ✓ Branches, query, Blackbox, policy, impact, import, pipelines, keys □ Versioned paths (<code>/api/v1</code>) ✓ Exact-origin CORS, branch-name check, body limits, timeouts ✓ Branches owned by their maker; others 404 (owner, admin, or main)	Unit; VM §1; v2 §0–§8; VM §1d
✓ K2	OpenAPI spec ✓ Served, Blackbox 2.0 endpoints documented ✓ Matches the code (import fields, status codes, auth routes, <code>/blackbox</code> aliases) ✓ Drift check in CI	Unit (drift check)
✓ K3	Python SDK ✓ Branches, query, Blackbox □ Import, pipelines, keys, agents □ Tests □ Publish to PyPI ✓ Contract tests against a live engine	Manual; VM sdks
✓ K4	TypeScript SDK ✓ Same client for Node and browsers □ Tests □ Publish to npm ✓ Compiles, and contract tests against a live engine	Manual; VM sdks
✓ K5	Agent Branch API ✓ A branch per agent, own non-superuser role ✓ Provenance (task, parent, session) ✓ DSN reachable from the host (gateway-routed, branch-scoped key) ✓ Cap, listing and TTL count suspended branches ✓ Superuser switch only in <code>-tags insecure</code> builds ✓ Agents and their keys belong to the caller	Unit; VM §5; v2 §1c, §7
✓ K6	MCP server ✓ Branches, <code>run_sql</code>, Blackbox tools ✓ <code>policy_check</code>, <code>execute_change</code>, <code>impact</code>, <code>diff</code> ✓ Requires an API key and acts as that account; a scoped key is held to its branch ✓ Client config & 16-tool reference (<code>docs/mcp.md</code>) ✓ Same ownership rule as the REST API	Unit; v2 §1c–§8

L Web console

11 of 11 done

ID	Feature · sub-features	Tested by
 L1	Embedded single-origin UI ✓ Served by the engine at :8080 ❑ Automated UI tests (whole area)	Manual
 L2	Login & signup ✓ Log in / sign up tabs, OAuth buttons ✓ Covered by the console's Playwright tests	Manual; VM ui
 L3	Dashboard ✓ Live tiles ✓ Branch table: create, suspend, resume, delete, copy DSN ✓ Covered by the console's Playwright tests	Manual; VM ui
 L4	SQL console & browser ✓ Schema sidebar, rows, structure, indexes ✓ Values shown as Postgres writes them (uuid, interval, inet, bytea) ✓ SQL editor ✓ System tables hidden by default, with a show button ✓ Runs as the signed-in user; one-run override switch ✓ Per-rule override for policy blocks 🟡 One statement per run (extended protocol)	Manual; API in v2 §6
 L5	Blackbox page ✓ Entries per branch with badges ✓ Table / risk / date filters, and "Clear filters" ✓ Verify button; a row expands to its statement and session ✓ Covered by the console's Playwright tests	v2 §5b; browser; VM ui
 L6	API keys page ✓ Create, copy, list, revoke ✓ A new key is shown inside its connection string (E3's next step) ✓ Covered by the console's Playwright tests	Manual; browser review 21 Sep; VM ui
 L7	Landing, Guide & Docs pages ✓ Landing, developer guide, reference ✓ Refreshed against the code (https, sslmode, MCP, Blackbox, import routes)	Type-check + build
 L8	Blackbox integrity page ✓ Verify, checkpoint, export, branch-before ✓ Used signed in; every result matched the CLI 🟡 Entry list needs a refresh to show a new change	Type-check + build; browser review 16 Sep
 L9	Policies page ✓ Rules, warn/block, custom rules, preview, evaluations ✓ Used signed in; matched the CLI, wrote only to the chosen branch ✓ Non-admin change refused with a readable message ✓ Who can override: list, grant, revoke	Type-check + build; API in v2 §6; browser review 16 Sep
 L10	App sidebar layout ✓ Every link opens its page, highlighted, with breadcrumb ✓ Collapse persists across a reload; theme toggle persists ✓ No console errors on any page ❑ Under-860px drawer (needs a manual look)	Type-check + build; browser review 16 Sep
 L11	Design system & FoxByte theme ✓ Tokens for colour, type, space, radius, elevation and motion; components use no raw values ✓ Ember palette in both themes; filled controls legible (<code>--accent-ink</code>) ✓ Fox mark from theme variables, wordmark from brand.json, favicon ✓ Console, Blackbox, Policies and API keys re-laid out; one table style ✓ Every page reviewed in both themes; console checked at 375px ❑ Not yet reviewed against a running engine with real data	Type-check + build; browser review 21 Sep

M Build, CI & releases

7 of 7 done

ID	Feature · sub-features	Tested by
 M1	Makefile targets ✓ build, vet, test, integration suites ✓ Suites run only in a throwaway test VM (guarded) ✓ release, web, WSL builds, feature-doc (both PDFs) ✓ <code>integration-sdks</code> , <code>integration-ui</code>	CI

✓	M2 Linux CI	✓ Build, vet, test, gofmt ✓ Release script tests, checklist sync check ✓ VM integration suites nightly in CI (one runner per suite)	□ Release tags gated on them ✓ Secret scanning, govulncheck, npm audit, UI type-check	CI
✓	M3 Windows CI	✓ Build, vet, test, Pester, installer encoding ✓ Guard tests read backslash paths, and generated files are pinned to LF (21 Sep)	□ End-to-end WSL install test	CI
✓	M4 Releases & GHCR image	✓ Binaries for 5 platforms + fox-verify ✓ SHA256SUMS merged safely between jobs ✓ Fails unless fox update's files are published	✓ Multi-arch Postgres image on GHCR ✓ Actions pinned by SHA; signed SHA256SUMS; build attestations; SBOM □ First signed release run on GitHub	Script tests in CI
✓	M5 WSL distro & ZFS builds	✓ Prebuilt distro image ✓ Downloaded and verified by the installer (seen on Windows) ✓ Preloaded image tagged as the engine runs it	✓ Imported on a fresh Windows install; preloaded images loaded ✓ Published zstd-compressed, with the gzip size measured beside it in every build log 🟡 zstd unpack on a real Windows install (unit-tested only); the gzip image was 685 MB	Actions artifacts; Pester; Windows run 16 Sep
✓	M6 Postgres image	✓ PostgreSQL 18 + wal-g for a fresh install; one image per supported major (16, 18) ✓ The image follows the data: an install on 16 keeps running 16 after an update ✓ A branch on the wrong major is refused before docker run, with the way out ✓ The major lives in one place, and a test holds the Dockerfile, release matrix and Windows preload to it	✓ All three suites pass on PostgreSQL 18.6: 108 + 277 + 51 checks (21 Sep) ✓ A command that moves an install from 16 to 18 (D6), and a portable backup to take first (D5) □ pgvector ✓ Base image pinned by digest; wal-g checked by SHA-256	Unit; all three suites on 18
✓	M7 Product name in one file	✓ brand.json → Go, shell, web and installer blocks, regenerated by make brand ✓ Stored names brand-free and frozen: a rename never migrates an install ✓ Retired env prefixes, state directory, VM names and installs still handled	✓ Guards: retired names anywhere, product names in stored SQL, durable names spelt twice 🟡 The sweep is text substitution: review its diff (it over-reached twice)	Unit (brand, gen, oldnames), in CI via go test

Planned or discussed, nothing built yet. When work starts, move the row into its area with the next id there.

ID	Item · scope	Related
☐	Multi-tenant projects & roles ☐ Projects / organisations ☐ Per-branch permissions (RBAC)	G1, H6
☐	Branch merge ☐ Apply one branch's schema changes to another	H18
☐	Scheduled backups & retention	D2
☐	Automatic HA failover ☐ Health detection, fencing ☐ Multi-host	D4
☐	Pipelines: scheduling, incremental / CDC extract, DAG view	J1
☐	More import sources: Cassandra, DynamoDB	I1–I8
☐	pgvector in the Postgres image	M6
☐	Cloud write-once anchors (S3 Object Lock etc.)	H10
☒	Release signatures — done 22 Sep 2026	A9, M4
☐	Hosted cloud tier	—
☐	Networked storage disaggregation (README roadmap; see the storage engine note on page 1)	C1, D1

Out of scope by decision

A custom storage engine (PostgreSQL's own engine is used, for native transaction speed), sharding, Raft / distributed transactions, multi-region, HTAP.